

الديناميكا الحرارية

Thermodynamics

Outline:

Chemical Thermodynamics: Definition, The Nature of Energy, Work and Heat, Energy Units, Heat Energy, System and Surroundings, Lowering the Energy of the system,

First Law of Thermodynamics: Internal Energy (E), Relating ΔE to Heat and Work, Heat and Enthalpy Changes (ΔH), Enthalpy (H), Enthalpy of Reaction, **Combustion of Hydrocarbons.**

Second Law of Thermodynamics: Spontaneous Processes and Entropy, Processes that increase the entropy of systems, Standard Entropy (S°), **Entropy and the Second Law of Thermodynamics**, The Effect of Temperature on Spontaneity, Entropy Changes in Chemical Reactions, **Heat Engine, Engine Efficiency, Practical Uses.**

Free Energy: Free Energy and Chemical Reactions.

Third Law of Thermodynamics, Zeroth Law of Thermodynamic.

► **CHEMICAL THERMODYNAMICS:**

Thermodynamics is the study of the effects of work, heat, and energy on a system.

Thermodynamics is only concerned with macroscopic (large-scale) changes and observations.

Chemical Thermodynamics deals with changes of energy and entropy associated with reactions. Entropy also explains whether a reaction under the given conditions is spontaneous or nonspontaneous.

The Nature of Energy:

Energy is defined as the capacity to do work or to transfer heat. Energy is the ability to cause various kinds of change, such as moving an object from one place to another, breaking a chemical bond, or removing an electron from an atom.

Work and Heat:

The work can be defined as force (F) multiplied by distance (d):

$$w = F \times d \quad (1)$$

A force is any kind of push or pull exerted on an object to another because of a difference in temperature.

In thermodynamics, work has a broader meaning that includes mechanical work:

$$w = -P\Delta V \quad (2)$$

where ΔV , the change in volume, is given by $V_f - V_i$.



(3)

$$\Delta V = V_{\text{final}} - V_{\text{initial}}$$

1. For gas expansion, $\Delta V > 0$, so $-P\Delta V$ is a negative quantity.
2. For gas compression (work done on the system), $\Delta V < 0$, and $-P\Delta V$ is a positive quantity.

The units for work done by or on a gas are liters atmospheres (L.atm). To express the work done in the more familiar unit of joules, we use the conversion factor:

$$1 \text{ L} \cdot \text{atm} = 101.3 \text{ J}$$

Example (1): A certain gas expands in volume from 2.0 L to 6.0 L at constant temperature. Calculate the work done by the gas if expands (a) against a vacuum and (b) against a constant pressure of 1.2 atm.

Solution:

- (a) Because the external pressure is zero, the work w done is:

$$\begin{aligned} w &= -P\Delta V \\ &= -0(6.0 - 2.0)\text{L} \\ &= 0 \end{aligned}$$

- (b) The external, opposing pressure is 1.2 atm, so:

$$\begin{aligned} w &= -P\Delta V \\ &= -(1.2 \text{ atm})(6.0 - 2.0) \text{ L} \\ &= -4.8 \text{ L} \cdot \text{atm} \end{aligned}$$

To convert the answer to joules, we write:

$$\begin{aligned} w &= -4.8 \text{ L} \cdot \text{atm} \times \frac{101.3 \text{ J}}{1 \text{ L} \cdot \text{atm}} \\ \therefore w &= -4.9 \times 10^2 \text{ J} \end{aligned}$$

Energy Units:

* The **SI unit** for energy is the **Joule (J)**. A joule is not a large amount of energy, and we often use kilojoules (kJ).

Traditionally, energy changes accompanying chemical reactions have been expressed in **calories**.

A **calorie (cal)** was originally defined as the amount of energy required to raise the temperature of one gram of water by one degree Celsius. It is now defined in terms of the joule:

$$1 \text{ cal} = 4.184 \text{ J (exactly)}$$

Heat Energy (Heat Capacity):

Energy can be transferred from one place to another in many forms, among which are electricity, sound, light, and heat. Heat is energy that is transferred from one place to another because of difference in temperature.



Systems with a high potential energy are less stable and more likely to undergo change than systems with a low potential energy. **Chemical reactions move spontaneously toward a low potential energy when possible.**

In general, energy can be converted from one form to another. **Energy is conserved.** The total energy lost by a system equals the total energy gained by its surroundings. Likewise, the total energy gained by a system equals that lost its surroundings. This important observation is known as the *law of conservation of energy*. Because it is the most elementary of thermodynamic concepts, it is also known as **the first law of thermodynamics.**

* First Law of Thermodynamics

Stated simply; **The total energy of the universe does not change;** that is, *energy can neither be created nor destroyed. The energy of the universe is constant.* This does not mean that the form of the energy cannot change. Indeed, chemical energies of a molecule can be converted to thermal, electrical or mechanical energies.

* Internal Energy (E):

طاقة الحركة + طاقة الوضع = الطاقة الداخلية

The total energy of a system is the sum of all the kinetic and potential energies of the component parts. The total energy is called **the internal energy**. Because there are so many types of motions and interactions, we cannot determine the exact energy of any system of practical interest. However, we can **measure the changes in internal energy that accompany chemical and physical processes.**

We define the change in internal energy, which we represent as ΔE (read "delta E"), as the difference between the internal energy of the system at the completion of a process and that at the beginning:

$$\Delta E = E_f - E_i \quad (7)$$

1. A positive (+) ΔE results when $E_f > E_i$, indicating that the system has gained energy from its surroundings.
النتيجة موجبة
2. A negative (-) ΔE is obtained when $E_f < E_i$, indicating that the system has lost energy to its surroundings.
النتيجة سالبة

In a chemical reaction, the initial state of the system refers to **the reactants**, and the final state refers to **the products**.

Initial state (Reactants) $\longrightarrow$ Final state (Products)

Relating ΔE to Heat and Work:

Any system can exchange energy with its surroundings in two general ways: as heat or as work. The internal energy of a system changes in magnitude as heat is added to or removed from the system, or as work is done on it or by it. We can use these ideas to write a very useful algebraic expression of the first law of thermodynamics. When a system undergoes any chemical or physical change, the accompanying change in its internal energy, ΔE , is given by the heat added to or liberated from the system, q , plus the work done on or by the system, w :

$$\Delta E = q + w \quad (8)$$

Heat added to the system is assigned a positive sign. Likewise, work done on the system is positive (Figure 1). Both the heat added to the system and the work done on it increase the internal energy (**positive**).

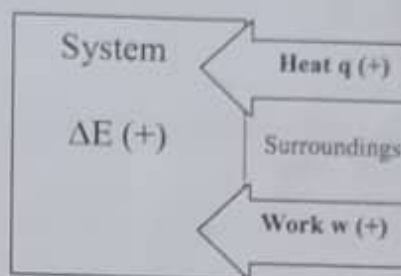


Fig. 1 Heat, q , absorbed by the system and work, w , done on the system.

On the other hand, both the heat lost by the system and the work done by the system on its surroundings are **negative**; they reduce the internal energy (Table 1).

Table. 1. The terms of heat and work.

Direction of Process	q	w
From system to surroundings	negative (-)	negative (-)
From surroundings to system	positive (+)	positive (+)

At constant volume, internal energy change (ΔE) is equal heat (q).

$$\Delta E = q_v \quad (9)$$

Example (3): If the system absorbed 50 J of heat and does 10 J of work on its surroundings. Calculate ΔE for the system.

Solution:

From the sign conventions for q and w , we have $q = + 50 \text{ J}$ and $w = - 10 \text{ J}$. By using the following equation, we have:

$$\Delta E = q + w$$

$$\Delta E = (+50 \text{ J}) + (-10 \text{ J})$$

$$\therefore \Delta E = 40 \text{ J}$$

Heat and Enthalpy Changes (ΔH):

When a reaction or physical change occurs in which the system absorb heat, we say the process is endothermic. In this case, heat flows into the system from its surroundings. A process that results in the evolution of heat is exothermic. That is, heat flows out of the system and into its surroundings (Fig.2).

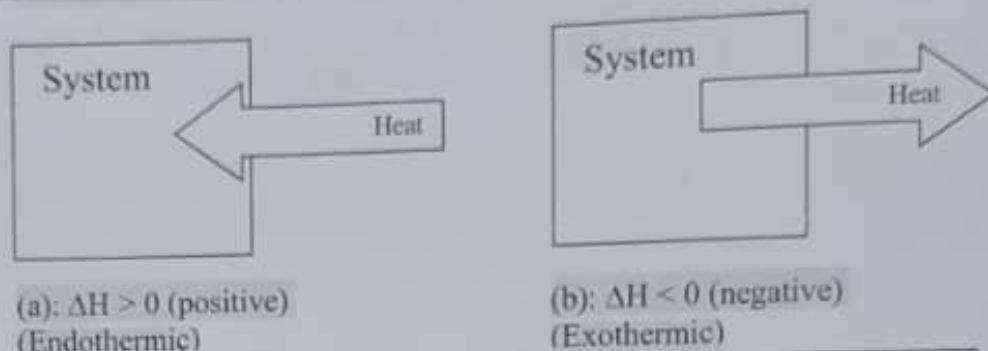


Fig.2. (a) If the system absorbs heat, $\Delta H > 0$
(b) If the system loses heat, $\Delta H < 0$.

ΔH represents the difference between the enthalpy of the system at the completion of a process and that at the beginning:

$$\Delta H = H_f - H_i \quad (10)$$

1. A **positive** ΔH results when $H_{\text{final}} > H_{\text{initial}}$, indicating that the system has **gained heat** from the surroundings (an endothermic process (Fig.2a))
2. A **negative** ΔH is obtained when $H_{\text{final}} < H_{\text{initial}}$, indicating that the system has **lost heat** to its surroundings (an endothermic process (Fig. 2b)).

Enthalpy (H):

Most physical and chemical changes including those in living systems, take place under the essentially **constant pressure** of the earth's atmosphere. Chemists use a special symbol, H , and a special name enthalpy in dealing with the heat absorbed or released under constant pressure. Enthalpy can not be measured, however, a change in the enthalpy, ΔH , is measured.



COMBUSTION

Combustion of Hydrocarbons:

Burning of hydrocarbon in presence of oxygen gas producing heat, carbon dioxide and water. **ALWAYS EXOTHERMIC!**

Rules and Examples:

1. complete combustion occurs when there is sufficient oxygen available
 - a. $C_xH_y + O_2 \rightarrow CO_2 + H_2O$
 - b. $C_xH_yO_z + O_2 \rightarrow CO_2 + H_2O$
 - c. $C_wH_yN_z + O_2 \rightarrow CO_2 + H_2O + NO_x$
 - d. $C_wH_yS_z + O_2 \rightarrow CO_2 + H_2O + SO_x$
 - e. complete combustion = blue flame
 - f. combustion reactions are also redox reactions because the oxidation states of carbon and oxygen change
 - i. Carbon is oxidized
 - ii. Oxygen is reduced
2. incomplete combustion occurs when there is insufficient oxygen available
 - a. $C_xH_y + O_2 \rightarrow CO + H_2O$
 - b. $C_xH_y + O_2 \rightarrow CO + C + H_2O$
 - c. incomplete combustion = smoky, yellow flame
 - d. Incomplete combustion forms carbon, carbon monoxide, and other hydrocarbons that didn't completely burn
 - e. The black soot is carbon and the yellow flame comes from glowing carbon atoms
 - f. CO is toxic for humans b/c it binds to hemoglobin in blood like oxygen – but unlike oxygen, it is never released and can cause suffocation if enough sites on the hemoglobin molecule are blocked.
 - g. Unburned hydrocarbons react with other molecules in the air (like NO₂) and make ozone which is also toxic for humans to breath.
 - h. Particulates in the air from unburned hydrocarbons can also impair breathing ability.



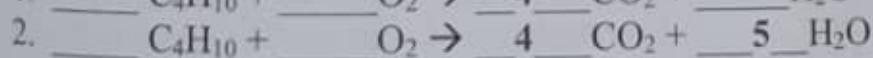
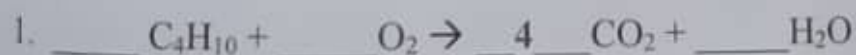
3. How to balance combustion reactions

- a. Balance the carbons
- b. Balance the hydrogens
- c. If you need an odd number of oxygens, double all coefficients

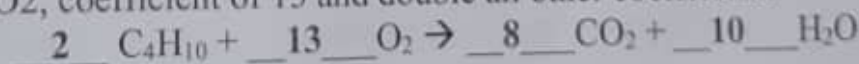


Example (5): Balance: $C_4H_{10} + O_2 \rightarrow CO_2 + H_2O$

Solution:



3. Need 13 oxygens – so give O₂, coefficient of 13 and double all other coefficients





* **Example (6):** Hydrocarbons are organic compounds composed entirely of hydrogen and carbon. A 0.1647 gram sample of a pure hydrocarbon was burned in a C-H combustion train to produce 0.4931 gram of CO_2 and 0.2691 gram of H_2O . Determine the masses of C and H in the sample and the percentages of these elements in this hydrocarbon.

Plan

Step 1: We use the observed mass of CO_2 , 0.4931 grams, to determine the mass of carbon in the original sample. There is one mole of carbon atoms, 12.01 grams, in each mole of CO_2 , 44.01 grams; we use this information to construct the unit factor:

$$12.01 \text{ g C} / 44.01 \text{ g CO}_2$$

Step 2: Likewise, we can use the observed mass of H_2O , 0.2691 grams, to calculate the amount of hydrogen in the original sample. We use the fact that there are two moles of hydrogen atoms, 2.016 grams, in each mole of H_2O , 18.02 grams, to construct the unit factor:

$$2.016 \text{ g H} / 18.02 \text{ g H}_2\text{O}$$

Step 3: Then we calculate the percentages by mass of each element in turn, using the relationship

$$\% \text{ element} = \frac{\text{g element}}{\text{g sample}} \times 100\%$$

Solution:

$$\text{Step 1: } \underline{\quad} \text{ g C} = 0.4931 \text{ g CO}_2 \times \frac{12.01 \text{ g C}}{44.01 \text{ g CO}_2} = 0.1346 \text{ g C}$$

$$\text{Step 2: } \underline{\quad} \text{ g H} = 0.2691 \text{ g H}_2\text{O} \times \frac{2.016 \text{ g H}}{18.02 \text{ g H}_2\text{O}} = 0.03010 \text{ g H}$$

$$\text{Step 3: } \% \text{ C} = \frac{0.1346 \text{ g C}}{0.1647 \text{ g sample}} \times 100\% = 81.72\% \text{ C}$$

$$\% \text{ H} = \frac{0.03010 \text{ g H}}{0.1647 \text{ g sample}} \times 100\% = 18.28\% \text{ H}$$

$$\text{Total} = 100.00\%$$

Second Law of Thermodynamics:

Spontaneous Processes and Entropy

A process is said to be *spontaneous* if (once started) it will *continue without outside help*. *Spontaneous processes may be fast or slow*. Thermodynamics enable us to predict whether a process will occur, but provides no information about how fast the process will occur. Consider the following spontaneous processes, but only in one direction:

Organic Chemistry

Outline:

The Nature of Organic Molecules, Alkanes and Their Isomers, Drawing Organic Structures, The Shapes of Organic Molecules, Naming Alkanes, Cycloalkanes, Reactions of Alkanes, Families of Organic Molecules: Functional Groups, Alkenes and Alkynes, Reactions of Alkenes and Alkynes, Aromatic Compounds and Their Reactions, Alcohols, Ethers, and Amines, Aldehydes and Ketones, Carboxylic Acids, Esters, and Amides, Synthetic Polymers

► The Nature of Organic Molecules:

The word **Organic** is one of the most overused in the English language. People use it as a derogatory term in phrases like **Don't eat that; it's not organic**. Of course, there is a precise scientific definition of the word. In science, Organic can be a biological or chemical term. In Biology it means anything that is living or has lived. The opposite is Non-Organic. In Chemistry, an Organic compound is one containing **Carbon** atoms. The opposite term is **Inorganic**.

Organic chemistry is the study of carbon, its compounds, and their properties. The only major carbon compounds considered inorganic are carbonates (for instance, calcium carbonate, alluded to above, which is one of the major forms of mineral on Earth) and oxides, such as carbon dioxide and carbon monoxide.

There are over six million organic compounds characterized, including the foods we eat, (made of carbohydrates, lipids, proteins, and vitamins), furs and feathers, hides and skins, and the organisms they came from. not to mention plastics, synthetic and natural fibers, dyes and drugs, insecticides and herbicides, ingredients in perfumes and flavoring agents, and petroleum products.

The name *organic* chemistry came from the word *organism*. Prior to 1828, all organic compounds had been obtained from organisms or their remains. The belief then was that the synthesis of organic compounds from inorganic compounds in the laboratory was impossible. All efforts had failed, and scientists became convinced that some "vital force" that living organisms had was necessary to make an organic compound. The synthesis of urea from inorganic substances in 1828 led to the disappearance of this *vital force theory*.

The Uniqueness of Carbon

The great number of carbon compounds is possible because of the ability of carbon to form strong covalent bonds to each other while also holding the atoms of other nonmetals strongly. Chains of carbon atoms can be thousands of atoms long, as in polyethylene.

This series of compounds are called **alkanes**. The lighter ones are gases and used as fuels. The middle ones (7 Carbons to 12 Carbons) are liquids used in petrol (gasoline). The higher ones are waxy solids. Candle wax is a mixture of alkanes.

After **Butane**, the names of these compounds are from the Greek for the number of Carbon atoms followed by the suffix **-ane**. So, **Decane** would have the formula $C_{10}H_{22}$.

Polythene is a very large alkane with millions of atoms in a single molecule. Apart from being flammable, alkanes are stable compounds found underground.

In the alkanes, all four of the Carbon valency bonds are taken up with links to different atoms. These types of bonds are called **single bonds** and are generally stable and resistant to attack by other chemicals. Alkanes contain the maximum number of Hydrogen atoms possible. They are said to be **saturated**.

The alkanes are not the only hydrocarbons.

An interesting phenomenon with organic molecules is called **isomerism**. Let us look at two compounds introduced earlier.

Isomerism is another reason why there are so many organic compounds. Isomers are compounds with identical molecular composition but their structures are arranged differently. Depending on how they are arranged, they may have similar or different properties.

Because the number of carbons per molecule increases as the compound gets more complex, the number of possible isomers for any given formula becomes very, very large.

Formula	Number of Isomers
C_8H_{18}	18
$C_{10}H_{22}$	75
$C_{20}H_{42}$	366,319
$C_{40}H_{82}$	6.25×10^{13} (approx.)

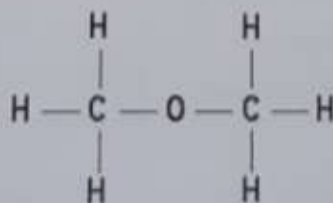


Examples of isomers:

Butane and 2-methyl propane both have the molecular formula C_4H_{10} .

Dimethyl Ether: $(CH_3)_2O$ and **Ethanol:** C_2H_5OH .

The first is a gas which will knock you out if inhaled. The second is common alcohol drunk in spirits. The two molecules are shown below.



Dimethyl Ether

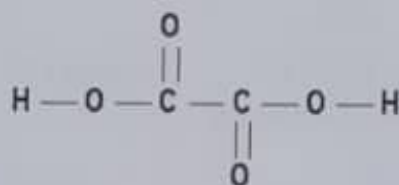
		group.		
$(C_nH_{2n+1})_2O$	Ethers	Ethers have an O atom attached to two hydrocarb on chains (or rings).	$(CH_3)_2O$ Dimethyl Ether a gas $(C_2H_5)_2O$ Diethyl Ether a liquid used as an anaesthetic)	<pre> H H H - C - O - C - H H H H H H - C - C - O - C - C - H H H H H </pre>
$(C_nH_{2n+1})_2CO$	Ketones	Ketones have a CO group attached to two hydrocarb on chains (or rings).	CH_3COCH_3 Dimethyl Ketone Also known as acetone: nail-varnish remover	<pre> H H H - C - C - C - H H O H </pre>
$C_nH_{2n+1}CHO$	Aldehydes	Aldehydes have a CHO group attached to a hydrocarb on chain (or ring).	$HCHO$ Formaldehyde preservative in labs CH_3CHO Acetaldehyde	<pre> H H - C - C - H H O H H H - C - C - C - H H H O </pre>
$C_nH_{2n+1}CO_2H$	Fatty Acids	Fatty Acids contain the CO_2H (or $COOH$) group attached to a hydrocarb on chain or ring.	HCO_2H Formic Acid in ant bites and stinging nettles CH_3CO_2H Acetic Acid vinegar $C_2H_5CO_2H$ Butyric Acid the rancid butter smell	<pre> H - C - O - H O H H - C - C - O - H H O H H H - C - C - C - O - H H H O </pre>



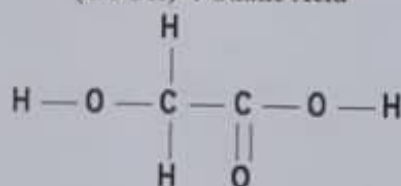
$\text{RCO}_2\text{R}'$ (R, R' are Hydrocarbon chains or rings).	Esters	Esters are similar to Fatty Acids except that the H in the COOH group is another hydrocarbon chain. They are usually very sweet smelling liquids used in perfumes.	$\text{CH}_3\text{CO}_2\text{CH}_3$ Methyl Methoate essence of pear drops	$ \begin{array}{c} \text{H} & & \text{H} \\ & & \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{C}-\text{H} \\ & & \\ \text{H} & \text{O} & \text{H} \end{array} $
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In the above examples, each molecule has a single functional group.

It is possible to have two or more functional groups on a molecule. These can be the same group (as in **Oxalic Acid** - a poison found in rhubarb leaves - which has two fatty acid groups) or different (as in **Hydroxymethanoic Acid** - which has a hydroxyl group and a fatty acid group):



$(\text{COOH})_2$: Oxalic Acid



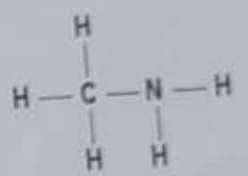
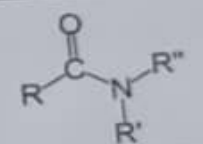
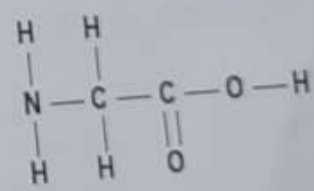
CH_2OHCOOH : Hydroxymethanoic Acid

The most famous compounds containing Carbon, Hydrogen and Oxygen are the **Carbohydrates**. An example is the common sugar, **Sucrose** ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$).

This shows how varied and complex even simple organic compounds can be. Sucrose has a pair of rings: one hexaganol, the other pentaganol. Each ring contains an Oxygen atom. The rings are joined by an Oxygen (Ether) link. The entire compound contains several Hydroxyl (OH) groups.

Adding Nitrogen

Many very important organic compounds contain **Nitrogen**. This produces more series of compounds.

General Formula	Series Name	Details	Examples	Structures
$C_nH_{2n+1}NH_2$	Amines	Amines have one or more of the Hydrogen atoms in Ammonia (NH_3) replaced by a Hydrocarbon chain or ring. Primary Amines have the formula RNH_2 Secondary Amines have the formula $RR'NH$ Tertiary Amines have the formula $RR'R''N$. (R, R', R'' are Hydrocarbon chains or rings).	CH_3NH_2 Methylamine a pungent, water soluble gas	
$R_1(CO)NR_2R_3$	Amides	An organic compound, such as acetamide, containing the $CONH_2$ radical.	CH_3CONH_2 Methyl amide	
$C_nH_{2n}NH_2COOH$	Amino Acids	Amino Acids have two functional groups: the amine (HN_2) group and the fatty acid ($COOH$) group. Amino Acids combine together to form proteins which are an important component of living organisms.	CH_2NH_2COOH Glycine the simplest amino acid.	



Drawing Organic Structures:

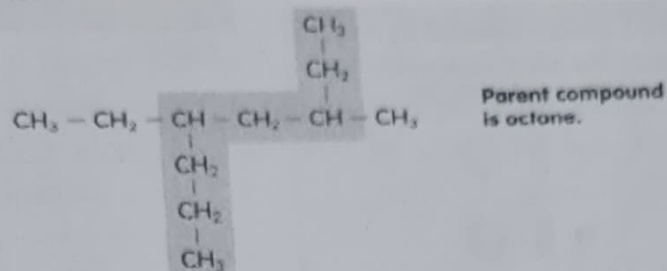
The structures of organic compounds are described by a molecular framework of carbon atoms on which substituents may be located at various points.

Structures can be represented in several ways, and the main steps for drawing organic structure are:

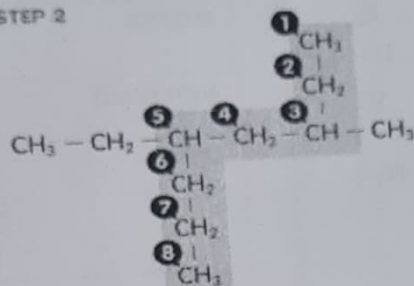
1. Line-angle structures - each intersection and endpoint is a C.
2. Carbon-carbon and carbon-heteroatom bonds are shown.

NAMING AN ALKANE MOLECULE

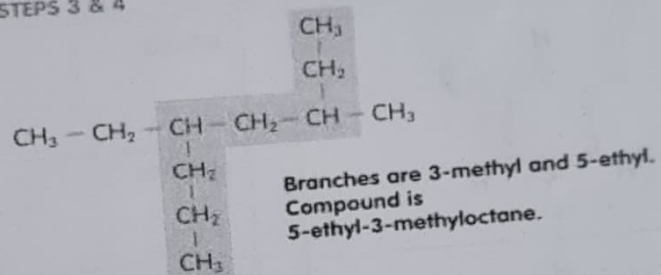
STEP 1



STEP 2



STEPS 3 & 4



Naming an alkane molecule